Sign Language Detection System Using YOLOv5

1. Introduction

Sign language is an essential medium of communication for people with hearing and speech impairments. However, a large portion of society does not understand sign language, creating a communication gap. To bridge this gap, this project focuses on developing a real-time sign language detection system using deep learning techniques.
The system leverages computer vision and object detection to recognize hand gestures and convert them into meaningful text output. By using advanced models like YOLOv5, the system ensures fast and accurate detection, making it suitable for real-time applications**.**

2. Abstract

This project presents an end-to-end Sign Language Detection System that uses the YOLOv5 object detection algorithm to identify and interpret hand gestures representing sign language. The system captures live video input, processes frames, and detects gestures in real time.
The dataset is collected using OpenCV and annotated using tools like Roboflow and Labeling.
The trained YOLOv5 model detects gestures and maps them to corresponding text outputs.
The system is deployed using Flask, enabling user interaction through a web interface.
This solution aims to assist communication between sign language users and non-signers, contributing to inclusivity and accessibility.

3. Tools and Technologies Used

Programming & Frameworks
- Python
- Flask (for deployment)

Libraries
- OpenCV (image processing & data collection)
- PyTorch (deep learning framework)
- NumPy & Pandas

Model
- YOLOv5 (You Only Look Once - Object Detection Model)

Annotation Tools
- Roboflow
- LabelImg

Development Tools
- Google Colab (training environment)
- VS Code

Cloud & Storage
- AWS S3 (model storage & deployment)

4. Steps Involved in Building the Project

Step 1: Data Collection

- Captured images of hand gestures using OpenCV.
- Ensured variations in lighting, angles, and backgrounds.
- Stored images in structured folders.

Step 2: Data Annotation

- Uploaded images to Roboflow.
- Annotated hand gestures with bounding boxes.
- Exported dataset in YOLOv5 format.

Step 3: Data Preprocessing

- Split dataset into training, validation, and testing sets.
- Performed augmentation (rotation, flipping, scaling).

Step 4: Model Training

- Used YOLOv5 pretrained weights.
- Trained model on custom dataset using Google Colab.
- Monitored metrics like accuracy, loss, and mAP.

Step 5: Model Evaluation

- Tested model on unseen data.
- Improved performance by tuning hyperparameters.

Step 6: Model Deployment

- Built a Flask web application.
- Integrated trained model for real-time predictions.
- Enabled webcam-based gesture detection.

Step 7: Integration with AWS

- Stored trained model on AWS S3.
- Loaded model dynamically for deployment.

5. Conclusion

The Sign Language Detection System successfully demonstrates the application of deep learning and computer vision in solving real-world accessibility problems. The use of YOLOv5 ensures high accuracy and real-time performance.

This project can be further enhanced by:

- Expanding the dataset for more gestures
- Converting gestures into speech output
- Developing a mobile application

Overall, the system contributes towards building an inclusive society by reducing communication barriers.